

Live Financial Data Extraction Pipeline

Technical Validation & Pipeline Report — Q2 2026

Prepared by: Keisy Fernanda
Role: Analytics Engineer / Data Analyst
Stack: Python (Requests, Pandas, JSON)

Executive Summary

This project focused on building an automated data ingestion pipeline capable of retrieving live financial market data from public REST APIs. The solution eliminated manual extraction processes and provided structured datasets ready for Business Intelligence and analytical consumption.

Business Problem

An investment analytics firm relied on manual downloads from financial websites, creating delays, operational risks, and data quality issues. A scalable and automated extraction mechanism was required.

Project Objective

Develop a Python-based pipeline that connects to public APIs, extracts real-time financial data, transforms raw JSON responses into structured datasets, and prepares outputs for BI ingestion.

Technology Stack

Python, Requests, Pandas, JSON Processing, Data Transformation, Error Handling, REST API Integration.

Engineering Techniques

HTTP GET requests, parameterized API calls, JSON normalization, schema standardization, exception handling, and automated export generation.

Pipeline Validation & Results

The pipeline successfully automated financial data extraction, replaced manual processes, and generated clean, schema-validated datasets suitable for reporting and analytics environments.

Business Impact

Reduced manual effort, improved data availability, minimized operational risks, and accelerated analytical decision-making through timely access to financial information.

Skills Demonstrated

API Integration, Data Engineering, Data Transformation, Python Development, Error Handling, Data Validation, Business Intelligence Preparation, Analytics Automation.

Conclusion

The project demonstrated the implementation of a reliable data ingestion architecture capable of transforming real-time API data into business-ready analytical assets, improving both efficiency and data quality.